THE SOUTH AFRICAN MATHEMATICS OLYMPIAD

Senior First Round 2023

Solutions

1. **Answer A**

   $10^{2023} \div 10^{2022} = 10^{2023-2022} = 10^1 = 10$.

2. **Answer D**

   Option (i) is false, since 2 is an even prime number, and option (ii) is false, since 8 is an even cube, which leaves (D) as the only remaining choice. Clearly option (iii) is true, since $2, 4, 6, \ldots$ form an infinite set of positive even integers, so (D) is the correct answer.

3. **Answer A**

   The arrival time in Dubai is $9{:}55 + 8{:}20 + 2{:}00 = 20{:}15$ the same day.

4. **Answer C**

   $\widehat{P} = 3\widehat{Q} = \frac{1}{2}\widehat{R}$, so $\widehat{Q} = \frac{1}{3}\widehat{P}$ and $\widehat{R} = 2\widehat{P}$. Also $\widehat{P} + \widehat{Q} + \widehat{R} = 180°$, so $\frac{10}{3}\widehat{P} = 180°$, giving $\widehat{P} = 54°$.

5. **Answer B**

   The diameter of the larger circle is equal to 18 (the height of the rectangle), so its area is $\pi \times 9^2 = 81\pi$.
   (Note that the smaller circle and the width of the rectangle are irrelevant.)

6. **Answer A**

   After three minutes, Sarah's speed drops to $75\% \times 60 = 45$ words per minute. Suppose the total time taken is x minutes. Then $3 \times 60 + (x - 3) \times 45 = 900$, so $x - 3 = (900 - 180) \div 45 = 16$, and $x = 19$.

7. **Answer B**

   $17x+51y = 85$ gives $17(x+3y) = 17\times 5$, so $x+3y = 5$, and $19x+57y = 19(x+3y) = 95$.

8. **Answer E**

   $\frac{1}{x} = \frac{1}{2}(\frac{1}{5} + \frac{1}{10}) = \frac{3}{20}$, so $x = \frac{20}{3}$.

9. **Answer E**

   Counting from today, Debbie's days off will be in $8, 16, 24, 32 \ldots$ days, which are all even numbers. Refilwe's will be in $1, 7, 13, 20, \ldots$ days, which are all odd numbers. Thus they will never be off on the same day.

10. **Answer C**

    In the factorization of a perfect square, all the prime factors have even exponents. It is clear that 2023 is not divisible by 2, 3, or 5. The next prime gives success:

$2023 = 7 \times 289$, and trying $11, 13, 17$ gives $289 = 17^2$. Thus $2023 = 7^1 \times 17^2$, which needs to be multiplied by another 7 to get an even exponent for 7.

11. **Answer D**

After 20 hours, Hippy has R20 $\times$ 2^{20}. If Happy gambles for x hours, then he will have R80 $\times$ 2^x. So $20 \times 2^{20} = 80 \times 2^x = 20 \times 2^{(x+2)}$. Thus $20 = x + 2$ and $x = 18$.

12. **Answer B**

Since a cube has six faces, and each edge is shared by two adjacent faces, the minimum number is $6 \div 2 = 3$. It is possible to choose three edges such that no two belong to the same face (and no two are parallel), which achieves the required result.

13. **Answer D**

The interior angles of an n-sided polygon add up to $(n-2)180°$, so $A\hat{B}C = 540/5 = 108°$. Thus $F\hat{B}C = 360° - 108° - 90° = 162°$. Since triangle BFC is isosceles, it follows that $B\hat{F}C = \frac{1}{2}(180° - 162°) = 9°$.

14. **Answer A**

The paths from A to B can be classified as follows:

via X only	via X and Y	via Y only
$2 \times 1 = 2$	$2 \times 1 \times 3 = 6$	$1 \times 3 = 3$

This gives a total of 11 paths, of which 8 pass through X, so the probability is $\frac{8}{11}$.

15. **Answer D**

The area cleaned is the difference between two sectors, the smaller of radius 18 cm and the larger of radius $42 + 18 = 60$ cm. Each sector makes up $\frac{160}{360} = \frac{4}{9}$ of a full circle. The area cleaned is therefore $\frac{4}{9}\pi(60^2 - 18^2) = 4\pi(20^2 - 6^2) = 1456\pi$ cm^2.

16. **Answer A**

The points $(a, b \pm c)$ are at a distance $2c$ apart on a vertical line, which can be regarded as the base of the triangle. The point $(-a, c)$ lies at a horizontal distance $2a$ (the height of the triangle) from them. The area of the triangle is, therefore, $\frac{1}{2}(2c)(2a) = 2ac$.

17. **Answer A**

Triangles ADF and CDE are similar, so $\frac{AD}{CD} = \frac{AF}{CE} = \frac{9}{11}$. Since $CD = AB = 12$, it follows that $AD = 12 \times \frac{9}{11} = \frac{108}{11}$.

Alternatively, the area of the parallelogram is given in two different ways by $AB \times AF = AD \times EC$. Therefore, $12 \times 9 = AD \times 11$ which gives $AD = \frac{108}{11}$.

18. **Answer C**

If light 1 turns green at time t seconds, then light 2 turns green at $t + 10$ seconds, light 3 at $t+20$ seconds, and so on. Finally, light 8 turns green at $t+70$ seconds, and all lights are green for 20 seconds until light 1 changes to yellow at $t + 90$ seconds.

19. **Answer A**

Let x, y, z denote the floor areas in square metres covered by exactly one, two, three layers, respectively. Then $x + y + z = 64$ (the floor area covered by the rugs) and $x + 2y + 3z = 150$ (the total area of the three rugs). Since $z = 28$, it follows that $x + y = 64 - 28 = 36$ and $x + 2y = 150 - 84 = 66$, from which we obtain $y = 66 - 36 = 30$ (and $x = 36 - 30 = 6$).
(Alternatively, this problem can be solved using formulas from Venn diagrams.)

20. **Answer D**

Suppose (h, t) represents (heads, tails) at any stage in the fight. After the knight's next sword stroke, there are the following possibilities:

	One head	Two heads	One tail	Two tails
Result	$(h-1+1, t)$	$(h-2, t)$	$(h, t-1+2)$	$(h+1, t-2)$
Change	$(0, 0)$	$(-2, 0)$	$(0, 1)$	$(1, -2)$

Cutting off one head serves no purpose, so suppose x, y, z represent the respective numbers of the other three cuts required to remove all three heads and tails, ignoring the order of the cuts for the moment. Then $3 - 2x + z = 0$ for the heads and $3 + y - 2z = 0$ for the tails, giving $z = 2x - 3$ and $y = 2z - 3 = 4x - 9$. Since $z \geq 0$ it follows that $2x - 3 \geq 0$, giving $x \geq 2$, and since $y \geq 0$ it follows that $4x - 9 \geq 0$, giving $x \geq 3$. With $x = 3$ we obtain $y = 3$ and $z = 3$ also, so the least total number of strokes is $x + y + z = 9$.
It is important to check that $h \geq 0$ and $t \geq 0$ at all stages of the fight. This can be achieved by repeating the sequence One tail, Two tails, One head, which has the following effect: $(h, t) \to (h, t+1) \to (h+1, t-1) \to (h-1, t-1)$. Three repeats of this sequence will take $(3, 3)$ to $(0, 0)$ while keeping $h \geq 0$ and $t \geq 0$ at all times. (The problem can be represented as a path from $(3, 3)$ to $0, 0)$ in the (h, t)-plane, staying in the first quadrant all the time.)
